

Smartphone Price Prediction Using Machine Learning

Machine Learning Project

by:

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Background & Business Context

- The Indian smartphone market is one of the fastest-growing consumer electronics markets.
- Prices vary across Amazon, Flipkart, Croma, Reliance Digital and Myntra.
- Machine learning helps estimate fair smartphone prices from hardware specifications.



Problem Statement

- Develop a regression model to predict smartphone prices.
- Input features: Brand, RAM, Storage, Battery, Rating and Platform.
- Target variable: Listed Price (INR).
- Dataset size: 32,000 smartphone listings.

Project Objectives

- Build an accurate regression model.
- Analyze the impact of RAM, Storage and Battery.
- Compare Linear Regression, Random Forest, XGBoost and CatBoost.
- Evaluate using MAE, RMSE and R^2 .
- Identify the most important features.

Dataset Overview

Feature	Type
 Platform	Categorical
 Brand	Categorical
 Product Name	String
 Rating	Float
 RAM	Categorical
 Storage	Categorical
 Battery	Categorical
 Price	Target

 Total Records: 32,000

 Features: 10

 Target: Price

 Range: ₹8,156–₹81,894



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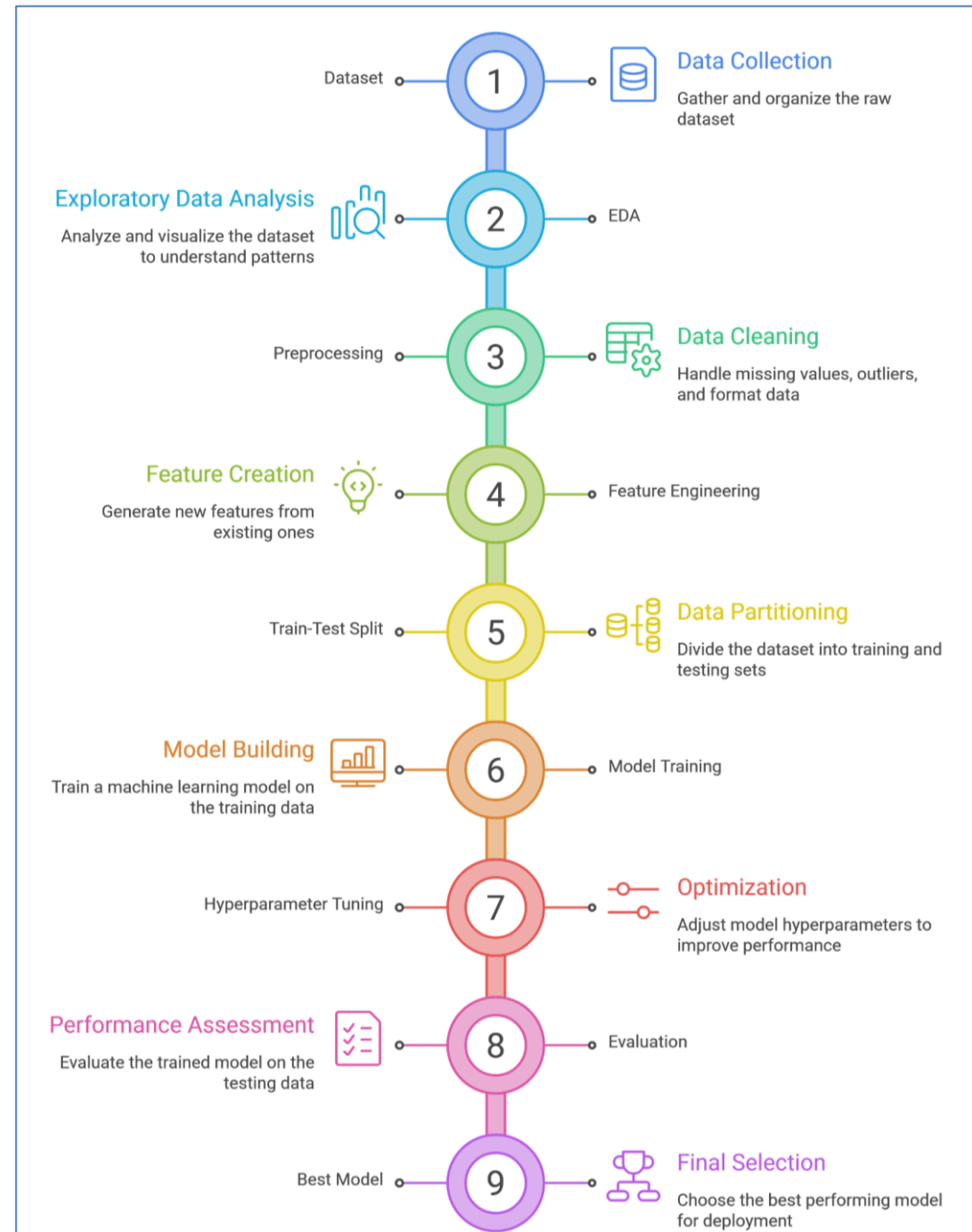
Target: Price



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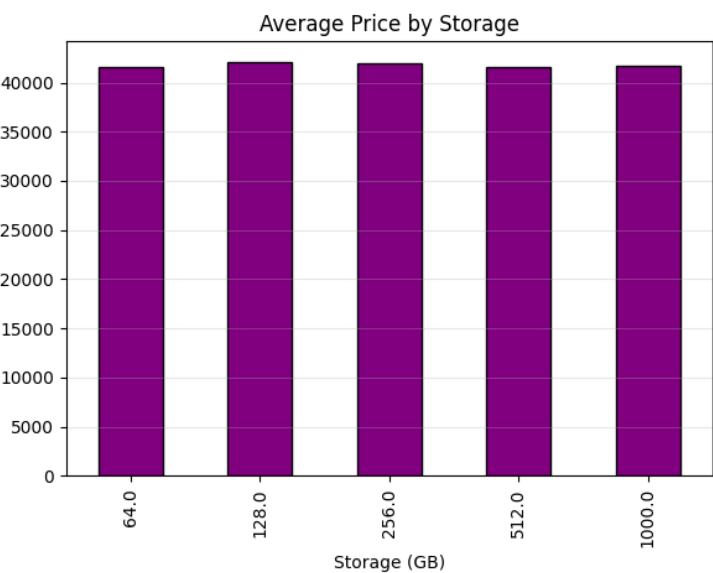
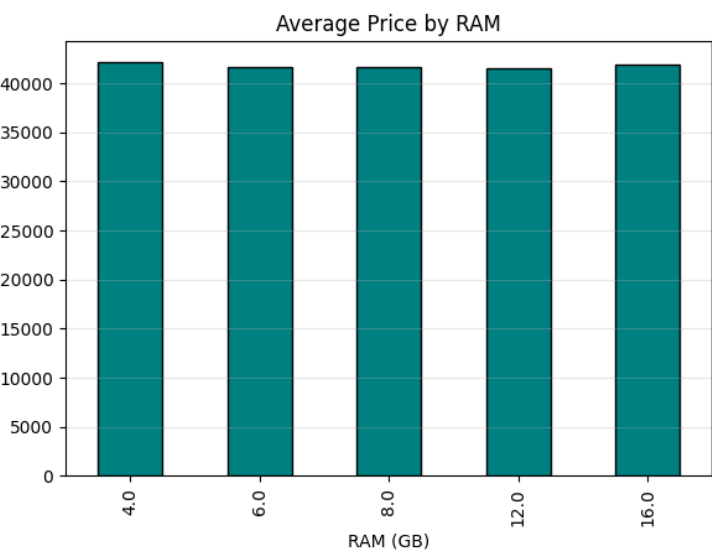
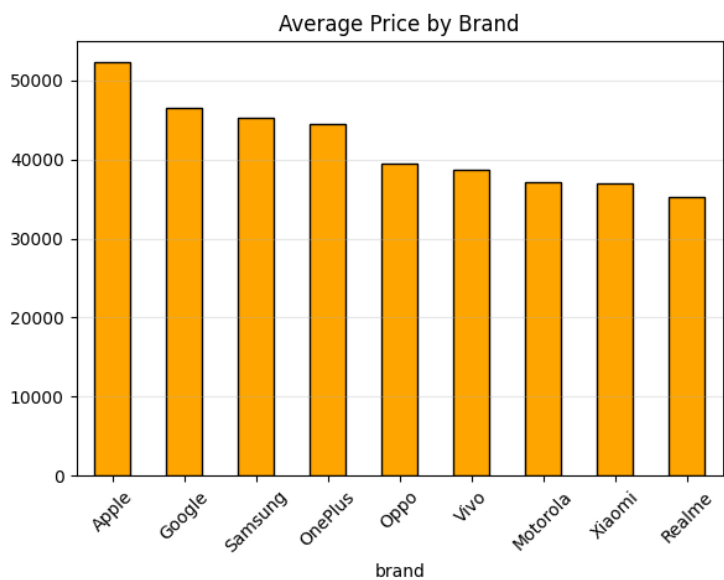
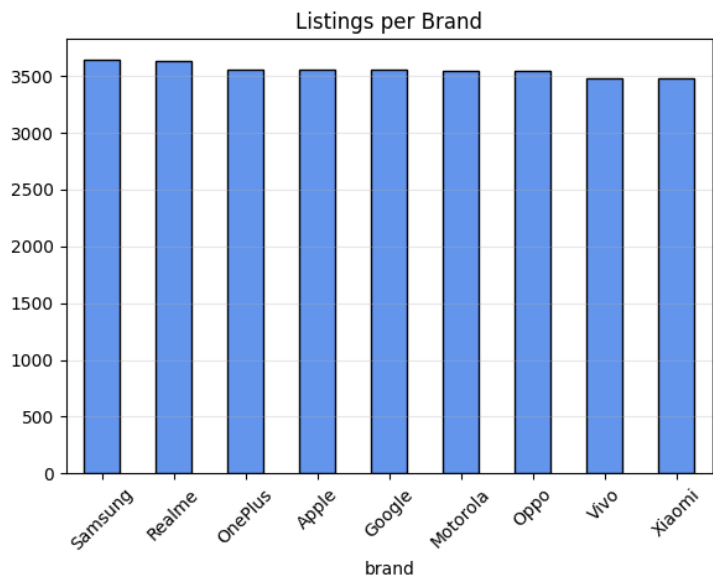
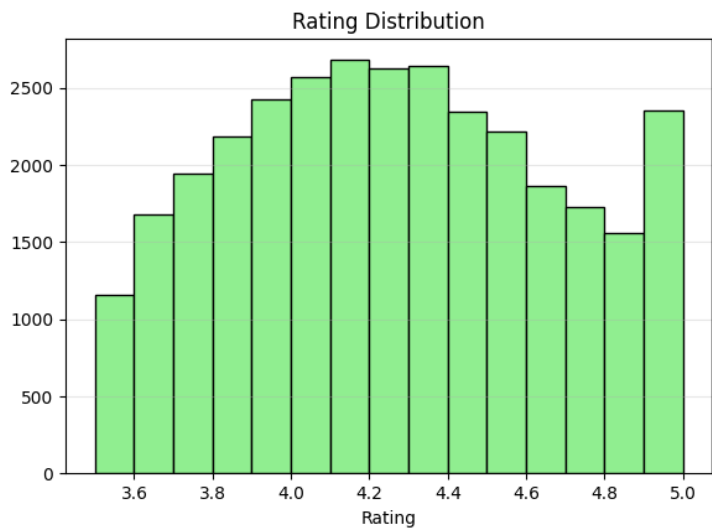
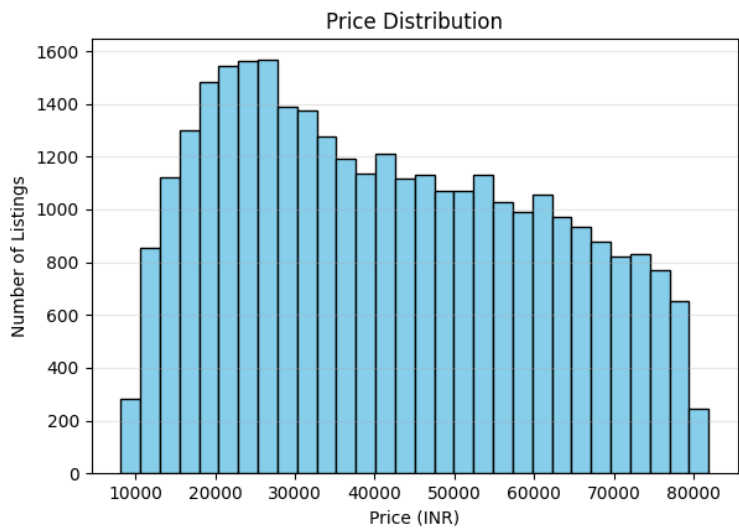
Methodology & Workflow -

- EDA
- Data Cleaning
- Feature Engineering
- Train-Test Split
- Model Development
- Hyperparameter Tuning
- Evaluation



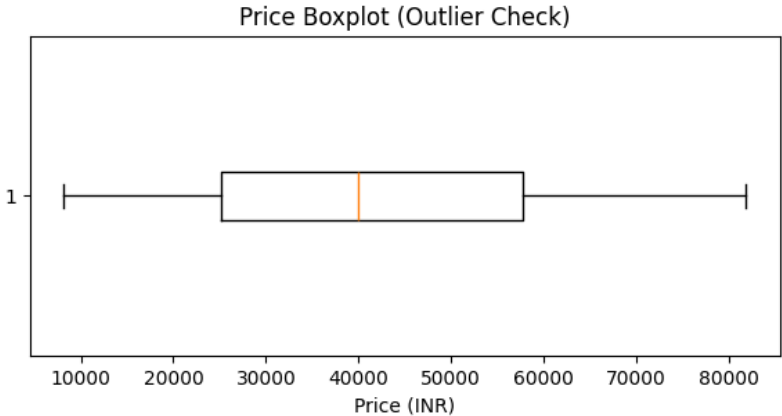
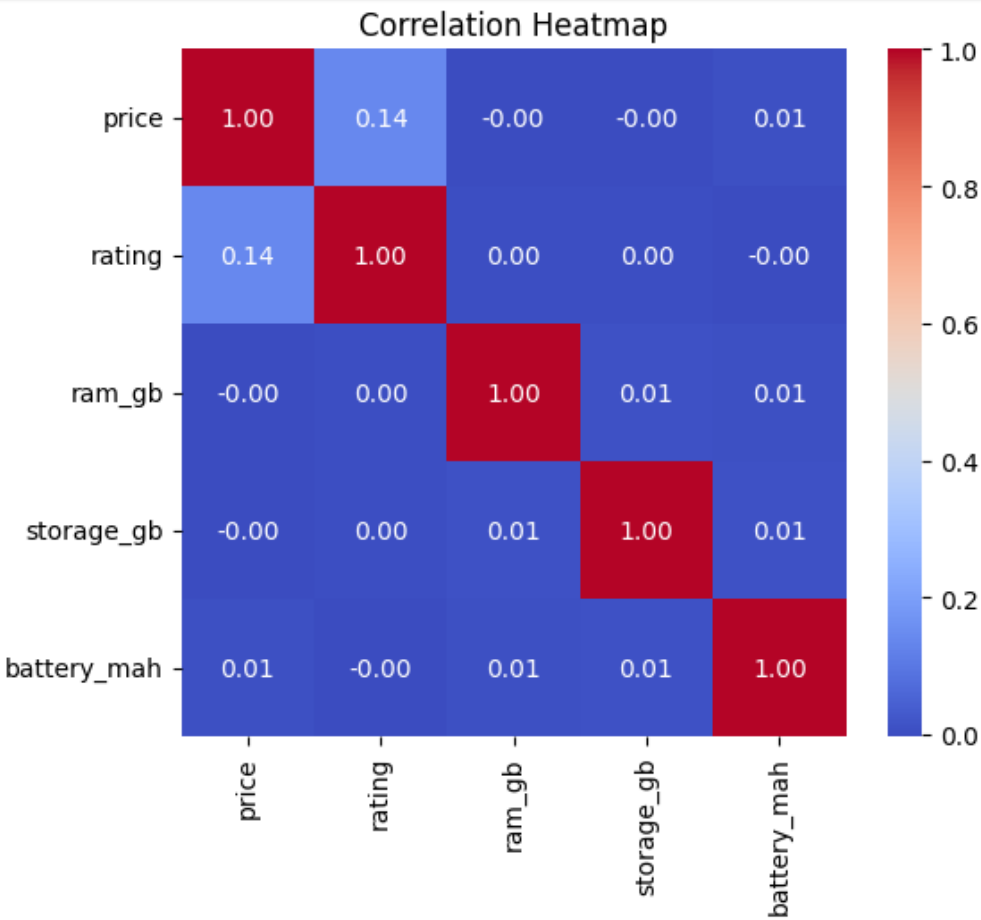
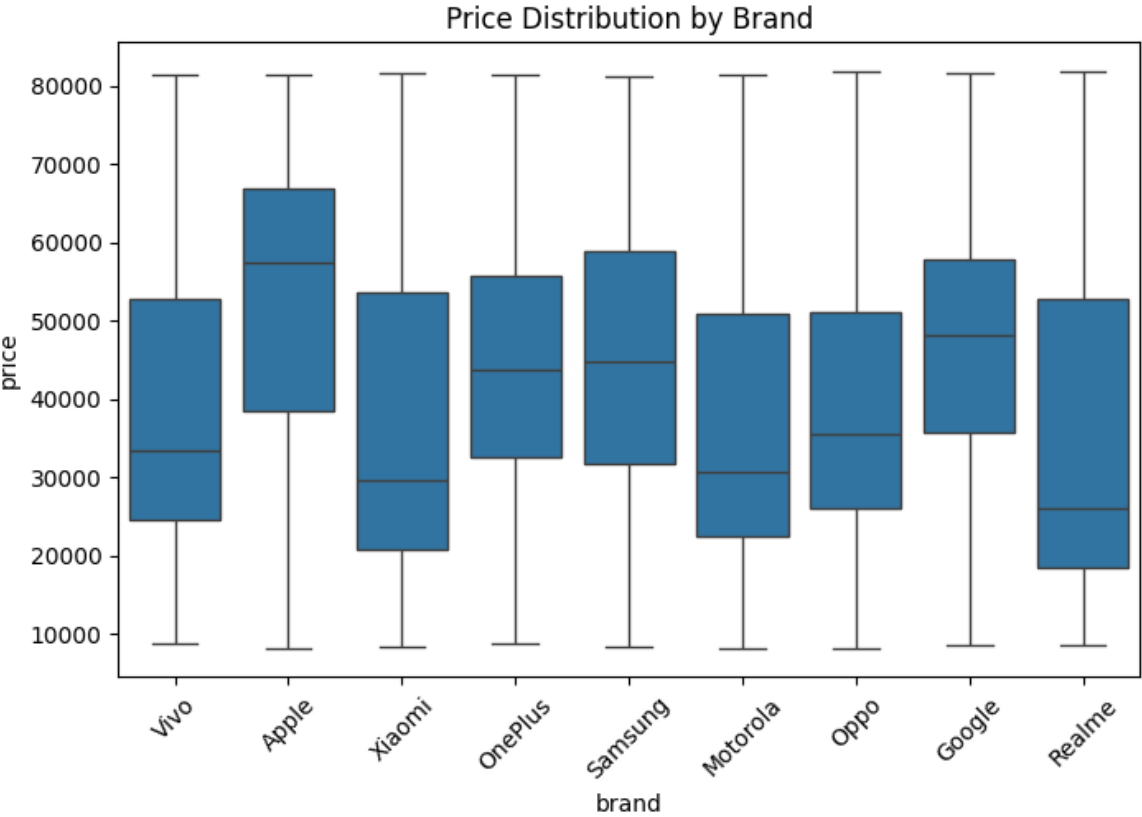
Exploratory Data Analysis

Exploratory Data Analysis (EDA)



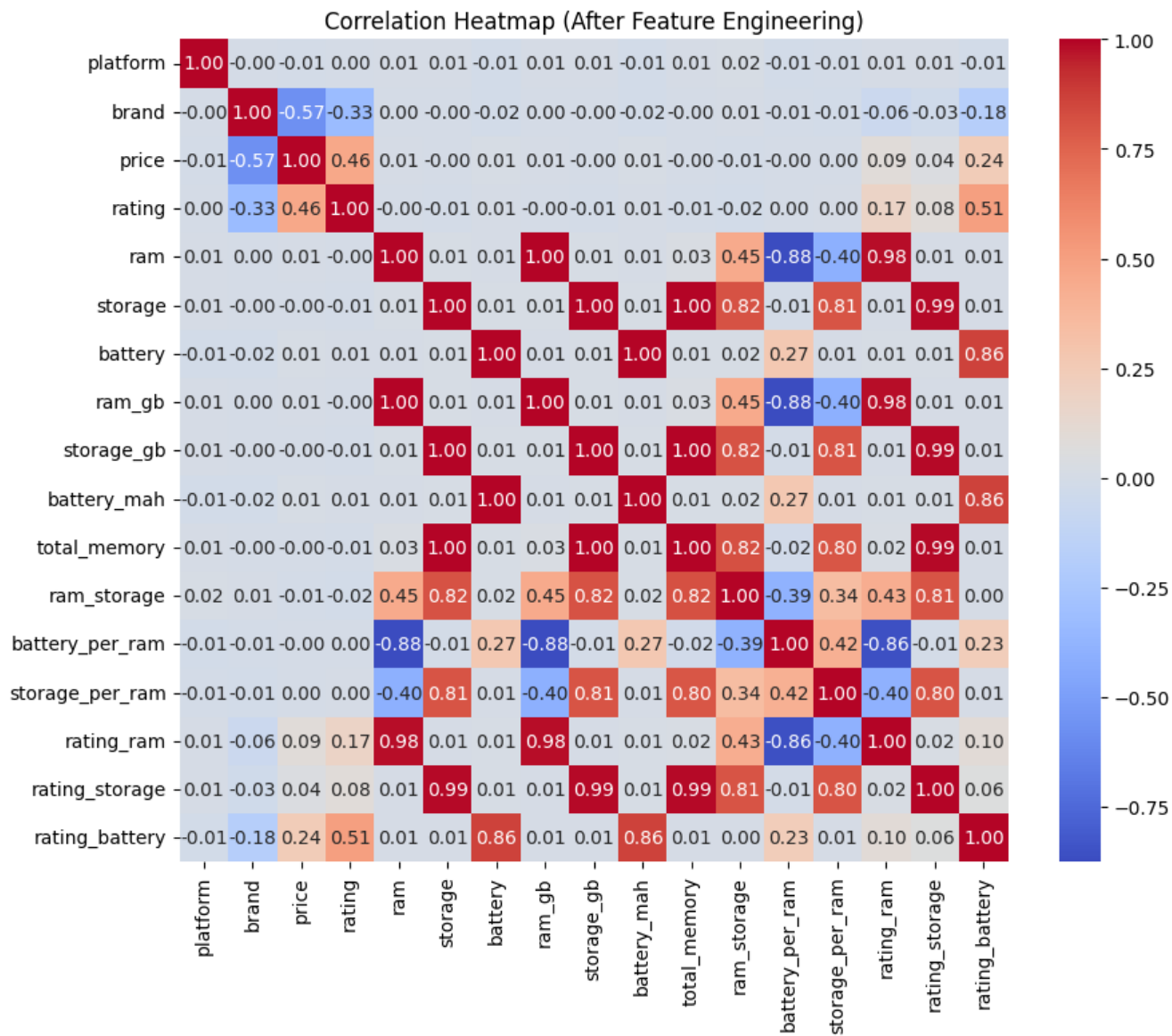
Exploratory Data Analysis on Original Data Set

- Price Distribution
- Brand-wise Price Analysis
- Correlation Heatmap (Observed relationships between hardware and price)

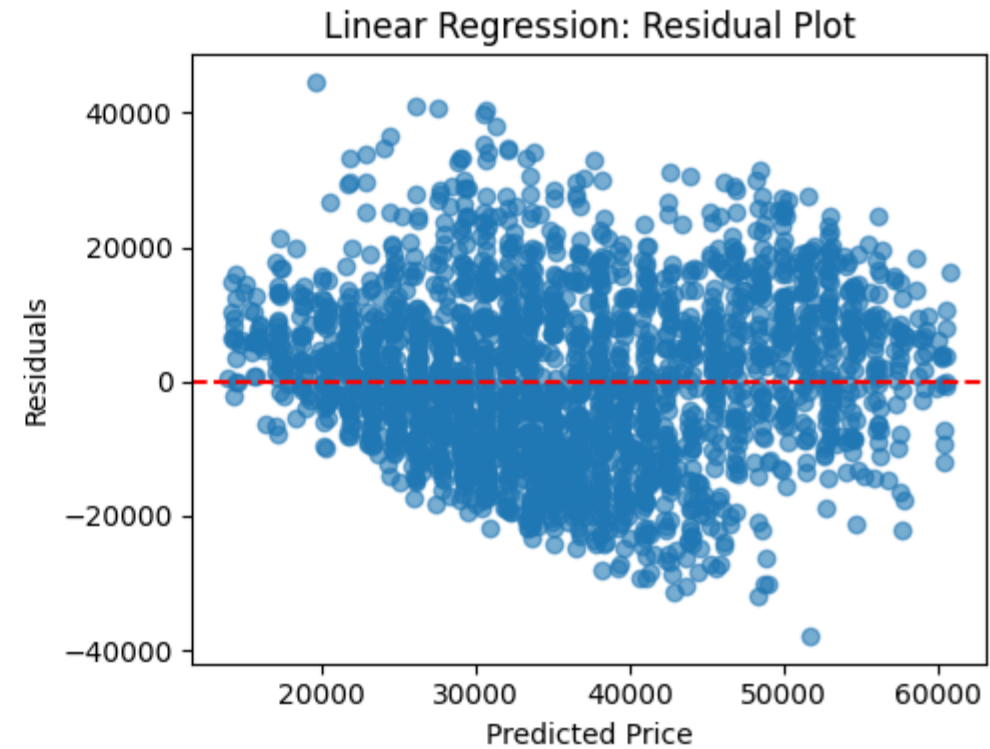
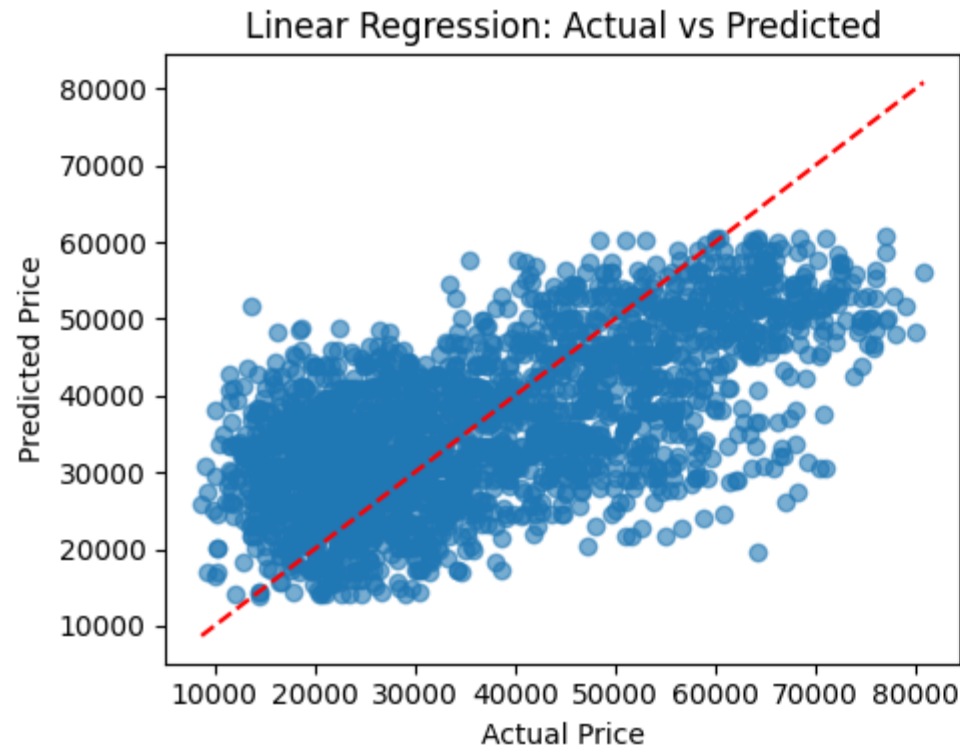


Feature Engineering on Synthetic Data Set-

Attribute	Value
Dataset Composition	20,000 original + 12,000 synthetic records (provided with the dataset)
Data Usage	Synthetic data is selected during preprocessing



Linear Regression -



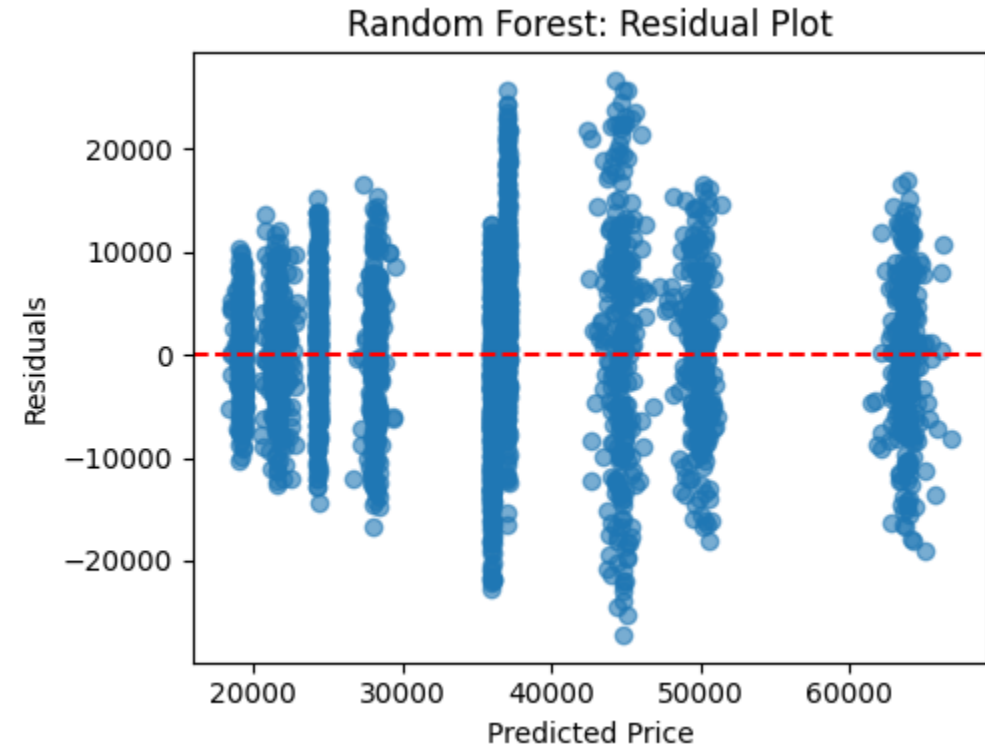
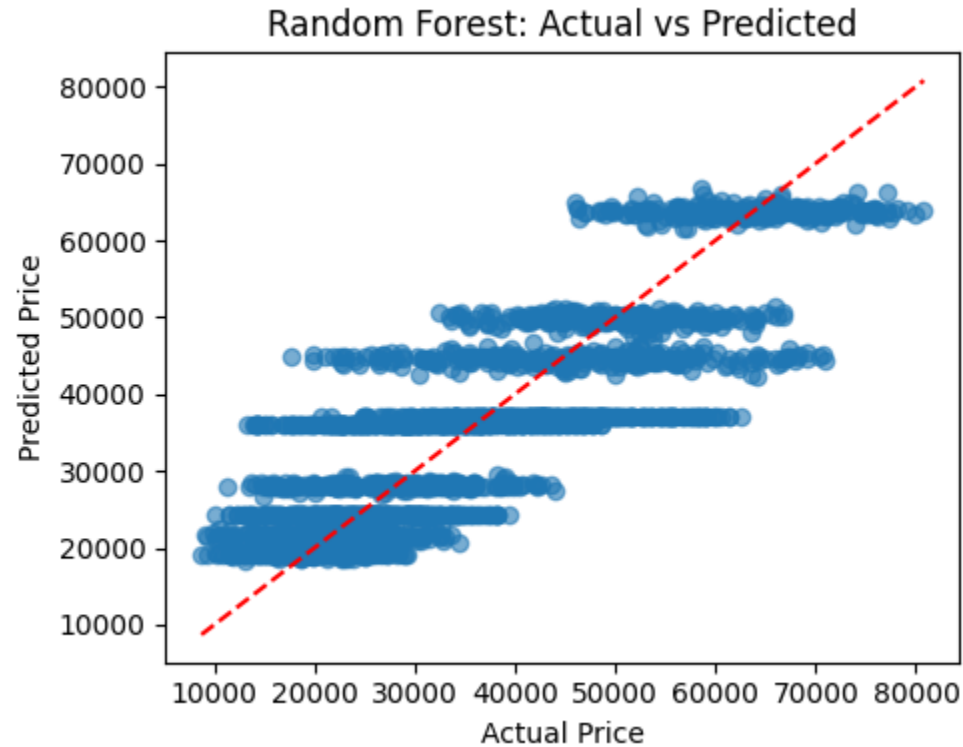
Linear Regression -

MAE : 10097.452113510668

RMSE: 12472.829871244943

R2 : 0.4124635612367401

Random Forest Regression -



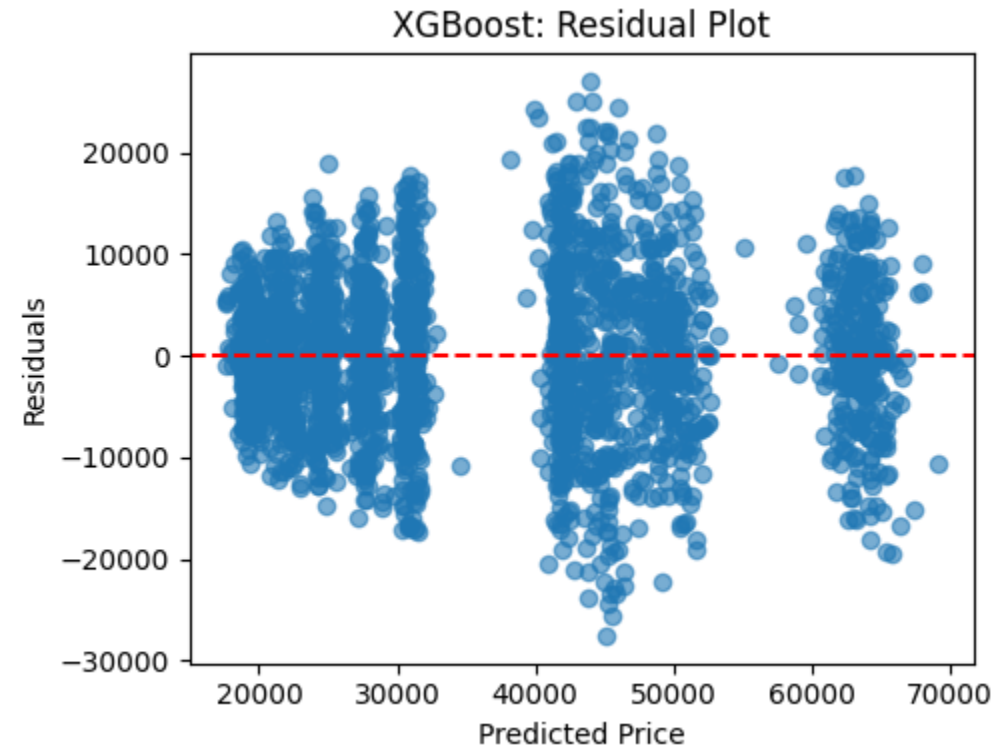
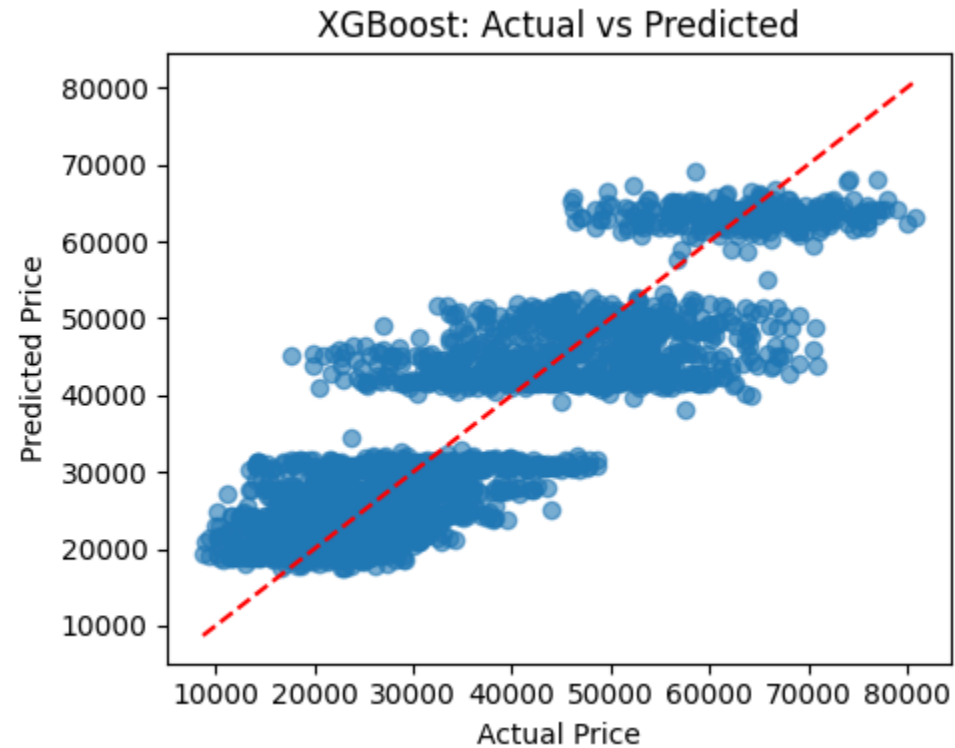
Random Forest -

MAE : 6351.767194440064

RMSE: 8174.5114241762385

R2 : 0.7476354089787355

XGBoost Regression -



XGBoost

MAE : 6107.2724609375

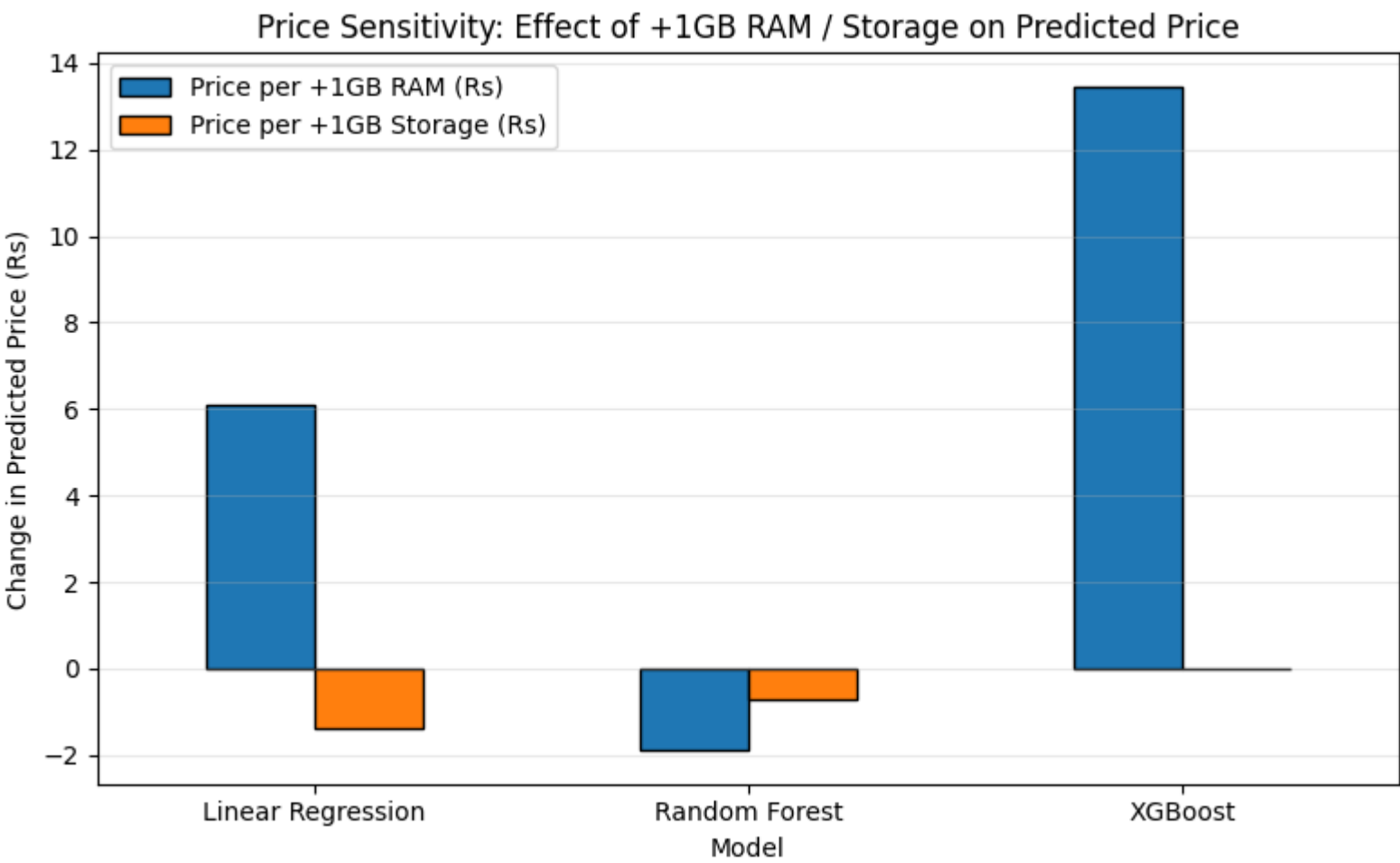
RMSE: 7792.19044941793

R2 : 0.7706895470619202

Model Comparison:

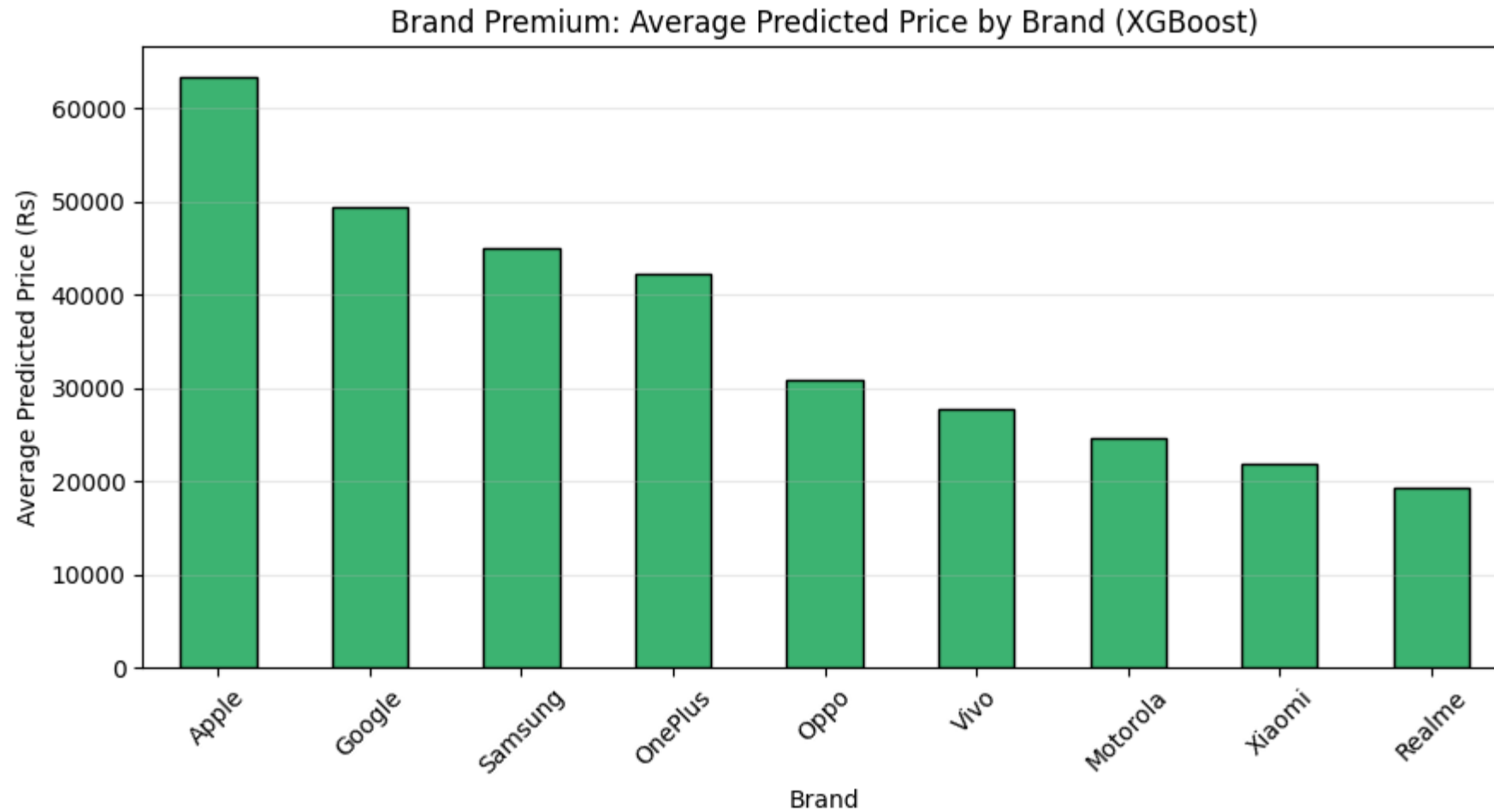
Model	MAE	RMSE	R ² Score
XGBoost	6,107.27	7,792.19	0.7707
Random Forest	6,351.77	8,174.51	0.7476
Linear Regression	10,097.45	12,472.83	0.4125

Price Sensitivity: Effect of



Model	Price per +1 GB RAM (₹)	Price per +1 GB Storage (₹)
Linear Regression	6.11	-1.37
Random Forest	-1.90	-0.72
XGBoost	13.46	0.00

Brand Premium



*Thank
you!*